

Transitive Inference with Exceptions

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Contents

1 Section 1: Derivation

Notes to remember

- Sam's original transitivity breaking document and before changing it in his paper had $f(x_j, x_k) = r_k - r_j$, where

$$r_j = \alpha'(b_{j-1} - b_j + \bar{h}(\delta_{jq} - \delta_{jp}))$$

and the margin was negative. This produces $D_{jp} - D_{jq}$. By reversing that definition so $f(x_j, x_k) = r_j - r_k$, we now have

$$r_j = \alpha'(b_j - b_{j-1} + \bar{h}(\delta_{jp} - \delta_{jq})).$$

This produces $D_{jq} - D_{jp}$.

Rank Representation

$$\tilde{r}_i(\alpha) = \frac{\sinh\left(\left(\frac{n+1}{2} - i\right)\lambda(\alpha)\right)}{\sinh\left(\left(\frac{n+1}{2}\right)\lambda(\alpha)\right) - \sinh\left(\left(\frac{n-1}{2}\right)\lambda(\alpha)\right)}$$

Given: Dual formulation $\implies y = Ka$ where $y = f(i, j) = r(i) - r(j)$
 Expression Definitions: $f(i, j) = \alpha'(b_i - b_{i-1} + b_{j-1} - b_j) + c\alpha'(\delta_{ip} + \delta_{jq} - \delta_{jp} - \delta_{iq})$

$$= r_i - r_j$$

$$= \tilde{r}_i = \frac{\sinh\left(\left(\frac{n+1}{2} - i\right)\lambda(\alpha)\right)}{\sinh\left(\frac{n+1}{2}\lambda(\alpha)\right) - \sinh\left(\frac{n-1}{2}\lambda(\alpha)\right)}$$

$$r_i = \alpha'(b_i - b_{i-1} + c(\delta_{ip} - \delta_{iq}))$$

$$\alpha'(b_i - b_{i-1}) = \tilde{r}_i + c\alpha'(D_{ip} - D_{iq})$$

$$D_{ij} = \begin{cases} \text{if } i = j : & -1 + 2 \left(\frac{\cosh\left((i \wedge j - \frac{1}{2})\lambda\right) \cosh\left((n - (i \vee j - \frac{1}{2}))\lambda\right) \tanh\left(\frac{\lambda}{2}\right)}{\sinh(\lambda n)} \right) \\ \text{if } i \neq j : & 2 \left(\frac{\cosh\left((i \wedge j - \frac{1}{2})\lambda\right) \cosh\left((n - (i \vee j - \frac{1}{2}))\lambda\right) \tanh\left(\frac{\lambda}{2}\right)}{\sinh(\lambda n)} \right) \end{cases}$$

$$\tilde{D}_{ij} = 2 \left(\frac{\cosh\left((i \wedge j - \frac{1}{2})\lambda\right) \cosh\left((n - (i \vee j - \frac{1}{2}))\lambda\right) \tanh\left(\frac{\lambda}{2}\right)}{\sinh(\lambda n)} \right)$$

$$\tilde{D}_{ij\text{constant}} = 2 \left(\frac{\tanh\left(\frac{\lambda}{2}\right)}{\sinh(\lambda n)} \right)$$

$$\tilde{D}_{ij\text{part}} = \cosh\left((i \wedge j - \frac{1}{2})\lambda\right) \cosh\left(\left(n - (i \vee j - \frac{1}{2})\right)\lambda\right)$$

$$c = \frac{1 + \left(\frac{\sinh\left(\left(\frac{n+1}{2} - q\right)\lambda\right) - \sinh\left(\left(\frac{n+1}{2} - p\right)\lambda\right)}{\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right)} \right)}{1 - 2\alpha' - 4\alpha' \left(\frac{\tanh\left(\frac{\lambda}{2}\right)}{\sinh(\lambda n)} \right) \sinh\left(\frac{q-p}{2}\lambda\right) D_{sim}}$$

$$c_{\text{denom2}} = c_{\text{denom2constant}} * c_{\text{denom2part}}$$

$$= 2\alpha' \frac{\tanh\left(\frac{\lambda}{2}\right)}{\sinh(\lambda n)} * \left(\sinh(\lambda) \sinh(\lambda n) - 2\sinh\left(\frac{n-p}{2}\lambda\right) D_{sim} \right)$$

$$D_{sim} = \left(\cosh\left(\left(q - \frac{1}{2}\right)\lambda\right) \sinh\left(\left(n - \frac{p+q}{2} + \frac{1}{2}\right)\lambda\right) \right. \\ \left. + \cosh\left(\left(n - p + \frac{1}{2}\right)\lambda\right) \sinh\left(\left(\frac{p+q}{2} - \frac{1}{2}\right)\lambda\right) \right)$$

$$c_{\text{num}} = 1 + \left(\frac{\sinh\left(\left(\frac{n+1}{2} - q\right)\lambda\right) - \sinh\left(\left(\frac{n+1}{2} - p\right)\lambda\right)}{\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right)} \right) \\ = \frac{c_{\text{numfinal}}}{c_{\text{numpart}}} = c_{\text{final}} \\ = \frac{(\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right) - \sinh\left(\left(\frac{n+1}{2} - p\right)\lambda\right) + \sinh\left(\left(\frac{n+1}{2} - q\right)\lambda\right))}{(\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right))}$$

Simplify: δ cancels with the case of D_{ij} when $i = j$

$$c \text{ simplifies with trig identity of } \tanh\left(\frac{x}{2}\right) = \frac{1 - \cosh(x)}{\sinh(x)}$$

$\alpha' D_{ij\text{constant}}$ cancels with $c_{\text{denom2constant}}$

f

$$\begin{aligned}
\text{Final: } \quad r_j &:= \alpha'(b_{j-1} - b_j + c(\delta_{jq} - \delta_{jp})) \\
r_j &= \tilde{r}_j + c\alpha'(\tilde{D}_{jq} - \tilde{D}_{jp}) \\
r_j &= \tilde{r}_j + c_{\text{final}} \left(\frac{\alpha'}{c_{\text{denomconstant}} * c_{\text{denom2part}}} \right) (D_{ij\text{constant}}) \left(\tilde{D}_{jp} - \tilde{D}_{jq} \right)_{\text{part}} \\
r_j &= \tilde{r}_j \\
&+ \frac{(\sinh(\frac{n+1}{2}\lambda) - \sinh(\frac{n-1}{2}\lambda) - \sinh((\frac{n+1}{2} - p)\lambda) + \sinh((\frac{n+1}{2} - q)\lambda)) (\tilde{D}_{jq} - \tilde{D}_{jp})_{\text{part}}}{(\sinh(\frac{n+1}{2}\lambda) - \sinh(\frac{n-1}{2}\lambda)) (\sinh(\lambda) \sinh(\lambda n) - 2 \sinh(\frac{q-p}{2}\lambda) D_{\text{sim}})} \\
r_j &= \tilde{r}_j + c_{\text{final}} \frac{(\tilde{D}_{jq} - \tilde{D}_{jp})_{\text{part}}}{(\sinh(\lambda) \sinh(\lambda n) - 2 \sinh(\frac{q-p}{2}\lambda) D_{\text{sim}})} \\
r_j &= \tilde{r}_j + \left(\frac{c_{\text{final}}}{c_{\text{denom2part}}} \right) (\tilde{D}_{jq} - \tilde{D}_{jp})_{\text{part}}
\end{aligned}$$

1.1 Simulation

r_diff_simulation.png

Figure 1: simulation from sympy TI trends proof